Geoeconomic fragmentation and its effects on Latin-American trade with a Dynamic bayesian extension*

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Abstract

This essay examines the effect of geoeconomic fragmentation on Latin America and the Caribbean's (LAC) trade flows. To investigate this question, the essay estimates a gravity model using bilateral trade data from the CEPII Gravity database and the Geopolitical Fragmentation Index developed by Fernández-Villaverde et al. (2024). The analysis is further complemented with indicators from the FLAR Economic Information System to classify countries according to trade openness and terms of trade vulnerability. The results show that higher geopolitical fragmentation is consistently associated with lower intraregional trade in LAC. A 0.1 unit increase in the fragmentation index is linked to a 2.9% reduction in intraregional LAC bilateral trade, with the effect becoming stronger after 2008. Free trade agreements appear to partially mitigate this negative effect, suggesting that institutionalized trade links can reduce exposure to global fragmentation. However, the results also indicate that some vulnerable LAC countries redirect trade toward external partners, including China, the United States, and the European Union. The bayesian estimates suggest a robust specification towards an overall policy recommendation to rely on regional free trade agreements to support trade flows. Overall, the essay argues that geoeconomic fragmentation has not contributed to creating regional cohesion. Instead, LAC is becoming increasingly dependent on external geopolitical blocs.

Keywords: *Geoeconomic fragmentation, gravity, Latin America & the Caribbean, trade*

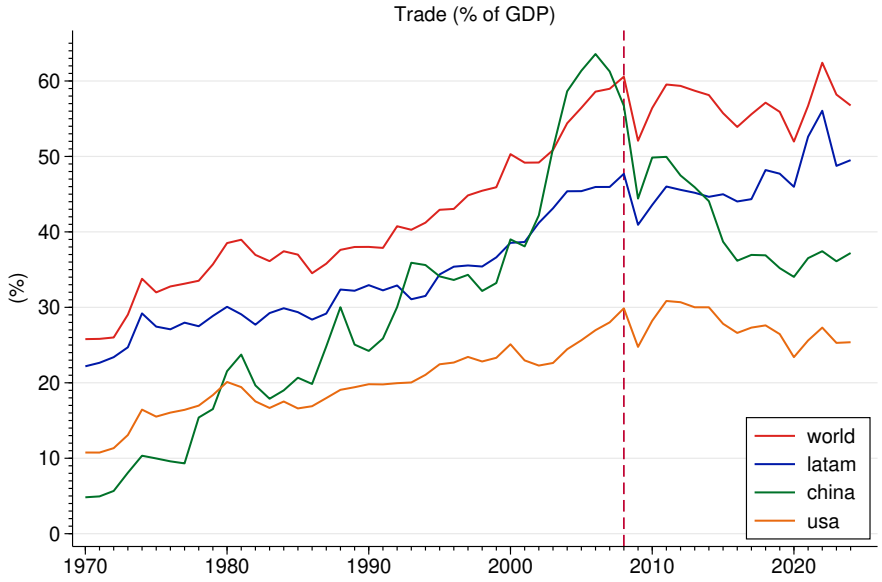
*I take full responsibility for any remaining errors in this work.

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1 Introduction

Global trade has quickly shifted away from the hyperglobalization trend towards slower, or even stagnant, growth in the years since the 2008 crisis, as shown in Figure 1. This shift has been referred to as slowbalization by Aiyar et al. (2023). In recent years, this phenomenon has been mainly driven by geoeconomic fragmentation, defined for this essay as the creation of economic blocs guided by geopolitics (Norring, 2024). Despite the potential gains of forming these associations for the parties involved, there are several risks linked with it, especially for developing countries benefiting from the gains of trade and liberalization. On the other hand, several authors argue that it presents an opportunity for unaligned blocs such as Latin America and the Caribbean (LAC). (Baur et al., 2025)

Figure 1: Global trade trends



Note: Figure displays the trade flow as a share of GDP for the world and three areas: LAC, China and USA.
Source: World Bank.

The channels affected by geoeconomic fragmentation include international trade, labor flows, foreign direct investment (FDI), capital flows, reserve currencies, debt management, and international cooperation. These channels are amplified by key mechanisms, such as uncer-

tainty and risk aversion, the previous degree of integration and reliance on foreign markets, and the role in global value chains (GVC's). (Norring, 2024) Firstly, a wide array of gains of trade has been documented for LAC. Even though the gains from trade have not been equally distributed, the era of liberalization drove economic growth for many years. Palma and Salas (2025) show that LAC's expansion of exports with China has reduced the sensitivity of terms of trade fluctuation and benefited greatly from the gains of trade. Chiquiar et al. (2016) demonstrates that when Mexico joined the North American Free Trade Agreement, wages and employment increased. Attinasi et al. (2025) analyses the effects of GVC decoupling and finds that consumer and producer prices increase in most countries, with a decrease in global trade due to reduced flows of intermediate outputs and greater reliance on domestic production.

For LAC, Machado Parente (2024) remarks on the potential reversal of trade gains via disruptions to trade and financial flows, shifts in remittances, migration, and international cooperation on climate change. On this note, the authors argue that trade integration within LAC could be a major mitigating tool against fragmentation. Gaps in infrastructure, governance, and human capital, along with heterogeneous technical barriers, could hinder integration of the region and reduce trade policy uncertainty. Further evidence on the opportunities for trade diversion is presented in Mayr-Dorn et al. (2026), which estimates the effects of the U.S.-China trade war and the diversion of Chinese exports to Vietnam on labor market outcomes and finds higher employment and lower informal employment in exposed districts. Chen et al. (2025) conduct a similar analysis for Mexico and conclude that there was an increase in labor demand among exporters exposed to US tariffs on China.

An open question in this context is the extent to which geoeconomic fragmentation is affecting trade among LAC countries. I use an empirical framework to approach this question, in which I estimate a gravity equation that accounts for global geopolitical fragmentation (as

proposed by Fernández-Villaverde et al., 2024) and countries' geopolitical distance (measured by Bailey et al., 2017). Then, I describe the main vulnerability channels that LAC faces due to geoeconomic fragmentation, using the Latin American Reserve Fund's Economic Information System (FLAR-SIE, by its Spanish acronyms) to enable consistent comparisons across several LAC countries using the main specification of the gravity model. The reason I chose a gravity approach is that it allows me to disentangle trends of trade linking them directly to geoeconomic fragmentation over time. Moreover, the SIE database allows me to split the sample via simple dummies to maintain parsimony in order to estimate differentiated marginal effects.

I find that a 0.1 unit increase in the Geopolitical Fragmentation Index is associated with a decline of 2.9% in intra-LAC bilateral trade in the baseline specification, and this effect becomes stronger reaching 5.2%. I also find that free trade agreements dissipate this effect. Then, using the SIE database to categorize vulnerable countries I identify a pattern of trade diversion away from intraregional trade. Finally, the bayesian approach suggests beneffits from trade agreements. These last results should be taken lightly since due to computational power, the number of draws is low relative to what we saw in class.

The main contribution of this essay is to connect the literature on geoeconomic fragmentation with a regional empirical analysis of LAC trade. Combining the Geopolitical Fragmentation Index by Fernández-Villaverde et al. (2024) with CEPII gravity data and SIE indicators, I show that fragmentation affects LAC through trade flows and the response is differentiated between countries. Section II covers the data used, section III estimates the model and presents results and limitations. Section IV discusses the bayesian extension. Finally, section V concludes and discusses policy implications

2 Data

I draw data from CEPII Gravity database. It gathers bilateral trade flows, from three distinct sources: IMF (DOTS), the UN (Comtrade) and the BACI database. It also describes the existence of trade agreements over time. Its data covers from 1948 to 2021. To have a balanced panel, the model data starts from 1975. I also gather data from the SIE, which homogenises key indicators from Latin American countries. It is important to note that some LAC countries do not have observations on certain periods. For this reason, the data used starts from 2003 till the second quarter of 2025. In the case of Venezuela I use the last observation, usually the second quarter of 2023. For the Geopolitical Fragmentation Index (GFI), I recover the dataset by Fernández-Villaverde et al. (2024). It contains the main index, combining trade, finance, mobility and political measures. These measures are also included in the dataset and are used for robustness.

3 Estimation and results

Equation (1) reflects my main specification.

$$trade_{ijt} = \beta_1 GFI_t \times \mathbf{1}\{i \wedge j \in LAC\} + \mu_{ij} + \delta_t + \varepsilon_{ijt} \tag{1}$$

On the left side, the variable $trade_{ijt}$ is the trade value (in current \$US thousands) between countries i and j in year t. On the right side, the variable GFI_t is the Geopolitical Fragmentation Index from Fernández-Villaverde et al. (2024), which captures the degree of global economic integration at time t. A higher value signals rising fragmentation. GFI_t is a time-specific measure that combines trade, finance, mobility, and political factors using a dynamic hierarchical factor model. I interact this variable with a dichotomous variable that

takes the value 1 if trade is LAC-intraregion, or if i and j are both LAC countries. Finally, I include country pairs and time fixed effects to capture time-invariant characteristics between the trading partners and general time-variant shocks, respectively. The main coefficient of interest is β_1, which captures the change in LAC's intraregional trade due to global fragmentation in column (1). I extend equation 1 to a simple model reflected in column (2) to estimate the case where either i or j are part of LAC.

In addition to this intraregional LAC effect, I consider an alternative specification -described in equation (2)- that explores the possibility of an interaction when a LAC country trades with a non-LAC country as an exporter or as an importer. Column (3) showcases the results.

$$trade_{ijt} = \beta_0 + \beta_1 GFI_t \times \mathbf{1}\{i \wedge j \in LAC\} + \beta_2 GFI_t \times \mathbf{1}\{i \in LAC\} + \beta_3 GFI_t \times \mathbf{1}\{j \in LAC\} + \mu_{ij} + \delta_t + \varepsilon_{ijt} \tag{2}$$

Additionally, I include a dummy interacted with the terms of equation (2) accounting for the case where a free trade agreement exists. Column (5) estimates these results. Moreover, to explain the structural shift of *slowbalization*, column (6) restricts the sample after 2008, while column (7) restricts it for before 2008.

Column 1 reports the main coefficient where the reference group is every other bilateral flow where both countries are not part of LAC. A 0.1 unit increase in the geopolitical fragmentation index is associated with approximately a 2.86% reduction in intra LAC bilateral trade. Column (2) gives us some insight into whether LAC reduces overall trade, or it substitutes away to other countries because the estimator becomes insignificant when either country involved is part of LAC. To further investigate these effects, column (3) reinforces the result that a 0.1 unit increase in the index results in a 2.88% reduction of intraregional trade

Table 1: PPML Estimates with Pair and Year Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
GFI_t × LAC - intraregion	-0.299***		-0.292***	-0.357***	-0.357***	-0.538***	-0.342***
	(0.058)		(0.058)	(0.093)	(0.093)	(0.127)	(0.115)
GFI_t × LAC - partner		0.036					
		(0.050)					
GFI_t × LAC - exporter			0.032		-0.077	-0.173	-0.121
			(0.086)		(0.119)	(0.172)	(0.075)
GFI_t × LAC - importer			0.106**		0.080	-0.087	0.179
			(0.050)		(0.065)	(0.114)	(0.111)
GFI_t × LAC - intraregion × FTA				0.116	0.130	0.084	0.268
				(0.109)	(0.109)	(0.151)	(0.166)
GFI_t × FTA				-0.091***	-0.105***	-0.073**	0.157***
				(0.025)	(0.024)	(0.035)	(0.061)
GFI_t × LAC - exporter × FTA					0.223*	0.414**	0.050
					(0.125)	(0.169)	(0.111)
GFI_t × LAC - importer × FTA					0.053	0.189	-0.051
					(0.097)	(0.128)	(0.128)
Observations	692353	692353	692353	692353	692353	400479	287749
Country-pair clusters	41771	41771	41771	41771	41771	39072	34502
Pseudo R-squared	0.980	0.980	0.980	0.980	0.980	0.989	0.989
Pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the results of econometric models that study the impact of global geopolitical fragmentation on LAC intraregion trade. The dependent variable is the level of bilateral trade. In all cases, the variable GFI_t is the Geopolitical Fragmentation Index from Fernández-Villaverde et al. (2024). The interaction terms are all dummy variables: LAC intraregion takes the value 1 if trade is between LAC countries, LAC exporter takes the value 1 if a LAC exporter trades, LAC importer takes value 1 if a LAC importer trades, and LAC partner takes value 1 if trade is made by a LAC country (either as an importer or exporter). FTA indicates if the exporter and the importer country have a free trade agreement. Column (6) restricts the sample to post 2008 while (7) restricts it to pre 2008. All estimates are obtained in a panel setting using the PPML estimator. Standard errors in parentheses Cluster-robust standard errors are clustered at the country-pair level. Pair and year fixed effects are included in all specifications. Constants are not reported. $^{*}p < 0.10,^{**} p < 0.05,^{***} p < 0.01$

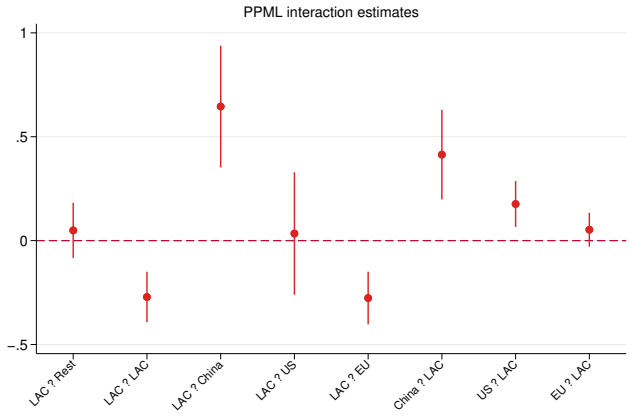
compared to all bilateral flows where neither country is LAC. Moreover, it gives us more insight into import competition for LAC as it seems imports rise with a shock to the index. However, the effect quickly dissipates in the following specification.

Next, column (4) and (5) delve into the role of free trade agreements. The Southern Common Market (MERCOSUR), Andean Community (CAN), CARICOM (Caribbean Community) or Central American Integration System (SICA) serve as facilitators of trade costs. Within our model the marginal effect of a shock to the GFI disappears when an agreement exists between two LAC countries. These findings support the arguments of Machado Parente (2024), supporting the elimination of trade barriers and strengthening infrastructure, governance, and human capital in order to integrate the region.

On the other hand, comparisons between columns (6) and (7) accentuate the shifting context of trade. The estimator ascends to a 5.24% reduction in intraregional trade when the index rises by 0.1 on the post 2008 sample. This result highlights the structural break of 2008 and confirms *slowbalization* as an important trend for LAC. Another result to note is that on both column (5) and (6), when LAC is an exporter, a shock to GFI increases flow to other countries when there is a free trade agreement. This estimate may reflect the shift to economic blocs over the last few decades of different LAC countries. To investigate this further I extend the analysis on equation (2) to include several economic blocs such as: European union (EU), United States (US) and China. The category of reference is the flow between countries not in EU, LAC, US or China.

Figure 2 categorizes flows between LAC and three other important economic blocs. A shock to GFI still reduces intraregional trade flows and European exports. Instead, it shifts them away to trade with China and some imports from the US. This narrative is consistent with findings by the strand of literature about trade diversion. Antras and Presbitero (2026) Palma and Salas (2025) Chen et al. (2025)

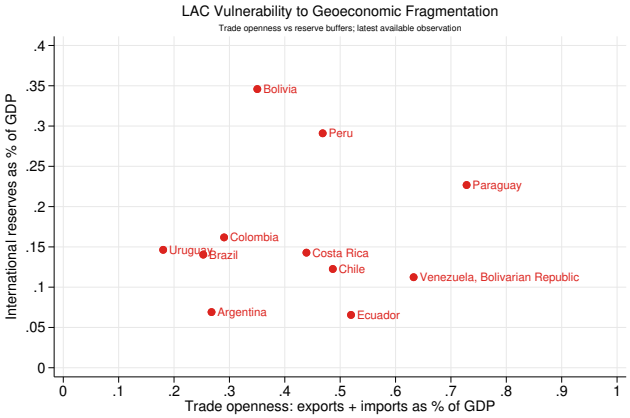
Figure 2: PPML estimates for economic blocs



Note: Figure displays only LAC related estimates)

Then, figure 3 plots trade openness against international reserves as a share of GDP for the LAC countries available in the sample. This figure provides a simple visual measure of exposure to geoeconomic fragmentation and how LAC is a diverse region with countries differently exposed to the effects of fragmentation. Countries located further to the right are more exposed through trade, while countries lower on the vertical axis have weaker external liquidity buffers. Thus, I extend the analysis by using four main specifications from Table 1 to divide the sample into two categories.

Figure 3



I split the sample using the SIE database. Constructing a trade openness measure by adding up exports and imports as a share of GDP, the sample is split into vulnerable countries, those

who are above the median of trade openness, and less vulnerable countries who are below the median. Within the more vulnerable category, the following countries are included: Chile, Bolivia, Costa Rica, Mexico, Paraguay and Venezuela. Respectively, the less vulnerable category is comprised of: Argentina, Brazil, Colombia, Ecuador, Peru and Uruguay. Table 2 reports the results.

Table 2 unveils a particular trend in which vulnerable countries have a lesser response to a shock to GFI compared to less vulnerable countries. However, when a vulnerable LAC country is an exporter or an importer the estimates are positive and significant. Moreover, when a free trade agreement is present the export flow diminishes to other countries. This pattern suggests a trend where vulnerable countries with high trade openness are shifting away from intraregional and trade agreements towards trade with other countries.

On the other hand, figure 4 shows the evolution of annual terms of trade variation in LAC. The blue line reports the regional average with the available countries in the given year, while the shaded area captures the interquartile range across countries. The figure highlights that LAC countries have been repeatedly exposed to external price shocks. More importantly, the dispersion across countries suggests that these shocks are not homogeneous among the region. This motivates the next analysis based on terms of trade vulnerability, since countries with weaker or more volatile external price conditions may respond differently to geopolitical fragmentation. The results hold when splitting the sample in terms of trade vulnerability. Within the more vulnerable category, the countries are: Chile, Ecuador, Mexico and Paraguay. Respectively, the less vulnerable category consists of: Argentina, Bolivia, Colombia, Peru and Uruguay.

For robustness, all models are estimated with other trade flow measurements provided by Comtrade and the FMI. The effects are more pronounced, suggesting that the ones presented here are lower bounds. Additional to that, Fernández-Villaverde et al. (2024) include other

Table 2: PPML Estimates by Trade Openness Vulnerability

	(1)	(2)	(3)	(4)
GFI_t × Vulnerable LAC intraregion	-0.389***	-0.382***	-0.476***	-0.476***
	(0.083)	(0.083)	(0.132)	(0.132)
GFI_t × Less vulnerable LAC intraregion	-0.184**	-0.178**	-0.169*	-0.169*
	(0.083)	(0.083)	(0.087)	(0.087)
GFI_t × Vulnerable LAC exporter		0.045		-0.238
		(0.119)		(0.297)
GFI_t × Less vulnerable LAC exporter		0.014		-0.031
		(0.105)		(0.111)
GFI_t × Vulnerable LAC importer		0.160**		0.185
		(0.065)		(0.170)
GFI_t × Less vulnerable LAC importer		0.042		0.019
		(0.044)		(0.050)
GFI_t × Vulnerable LAC intraregion × FTA			0.166	0.180
			(0.150)	(0.150)
GFI_t × Less vulnerable LAC intraregion × FTA			0.005	0.019
			(0.135)	(0.135)
GFI_t × FTA			-0.091***	-0.105***
			(0.025)	(0.024)
GFI_t × Vulnerable LAC exporter × FTA				0.398
				(0.294)
GFI_t × Less vulnerable LAC exporter × FTA				0.052
				(0.128)
GFI_t × Vulnerable LAC importer × FTA				0.004
				(0.173)
GFI_t × Less vulnerable LAC importer × FTA				-0.157
				(0.123)
Observations	692353	692353	692353	692353
Country-pair clusters	41771	41771	41771	41771
Pseudo R-squared	0.980	0.980	0.980	0.980
Pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

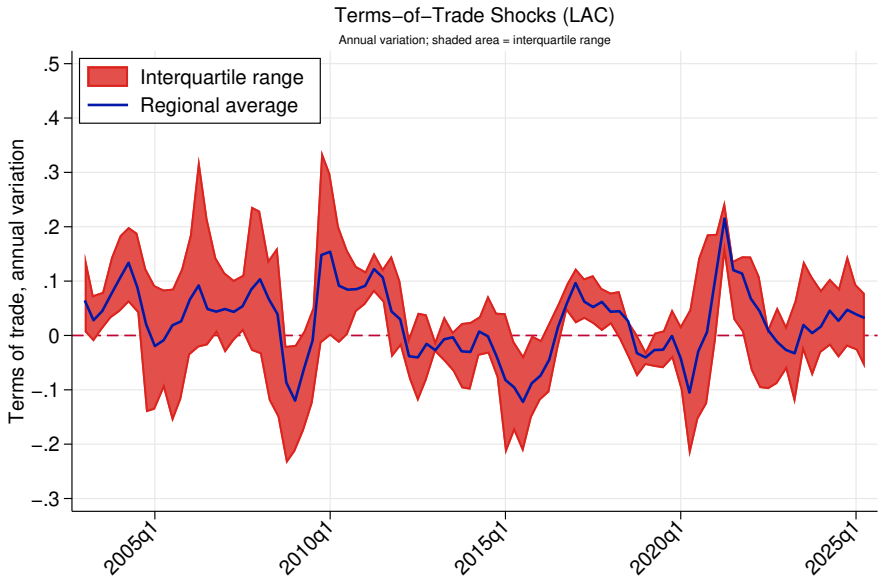
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Table 3: PPML Estimates by Terms-of-Trade Vulnerability

	(1)	(2)	(3)	(4)
GFI_t × Vulnerable LAC intraregion	-0.273***	-0.267***	-0.267***	-0.267***
	(0.064)	(0.064)	(0.100)	(0.100)
GFI_t × Less vulnerable LAC intraregion	-0.321***	-0.314***	-0.456***	-0.456***
	(0.090)	(0.090)	(0.149)	(0.150)
GFI_t × Vulnerable LAC exporter		0.169***		0.430**
		(0.035)		(0.207)
GFI_t × Less vulnerable LAC exporter		-0.128		-0.183
		(0.134)		(0.140)
GFI_t × Vulnerable LAC importer		0.220***		0.411***
		(0.050)		(0.113)
GFI_t × Less vulnerable LAC importer		-0.015		-0.042
		(0.052)		(0.058)
GFI_t × Vulnerable LAC intraregion × FTA			0.013	0.027
			(0.118)	(0.118)
GFI_t × Less vulnerable LAC intraregion × FTA			0.233	0.248
			(0.179)	(0.179)
GFI_t × FTA			-0.091***	-0.105***
			(0.025)	(0.024)
GFI_t × Vulnerable LAC exporter × FTA				-0.247
				(0.208)
GFI_t × Less vulnerable LAC exporter × FTA				0.176
				(0.134)
GFI_t × Vulnerable LAC importer × FTA				-0.211*
				(0.122)
GFI_t × Less vulnerable LAC importer × FTA				-0.097
				(0.118)
Observations	692353	692353	692353	692353
Country-pair clusters	41771	41771	41771	41771
Pseudo R-squared	0.980	0.980	0.980	0.980
Pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: This table reports the results of econometric models that study the impact of global geopolitical fragmentation on LAC intraregion trade. The dependent variable is the level of bilateral trade. In all cases, the variable GFI_t is the Geopolitical Fragmentation Index from Fernández-Villaverde et al. (2024). The interaction terms are all dummy variables: LAC intraregion takes the value 1 if trade is between LAC countries, LAC exporter takes the value 1 if a LAC exporter trades, LAC importer takes value 1 if a LAC importer trades, and LAC partner takes value 1 if trade is made by a LAC country (either as an importer or exporter). FTA indicates if the exporter and the importer country have a free trade agreement. Standard errors in parentheses. Cluster-robust standard errors are clustered at the country-pair level. Pair and year fixed effects are included in all specifications. Constants are not reported. Vulnerable countries are those with below-median average Terms of Trade in the FLAR sample. $^{*}p < 0.10,^{**}p < 0.05,^{***}p < 0.01$

Figure 4



measurement they use for their index. All models remain relatively the same, differing by a few points in some estimates, but remaining significant.

The results above show consistently that higher geopolitical fragmentation is associated with lower intraregional LAC trade, especially after 2008. This also shows that *slowbalization* has weakened regional trade linkages rather than reinforcing them. At the same time, the extended models reveal that this effect is not uniform: free trade agreements partly cushion the decline, vulnerable countries appear to redirect trade outside the region, and flows with major blocs such as China, the United States, and the European Union suggest a process of trade diversion rather than simple trade contraction. If LAC wants to avoid becoming passive recipients of external bloc formation, regional integration, infrastructure, governance, and trade facilitation with free trade agreements and similar instruments that reduce trade costs need to become more central to a development strategy to integrate the region further.

Several limitations should be acknowledged. First, the Geopolitical Fragmentation Index is a global time varying measure rather than a bilateral indicator of geopolitical distance. Considering the diverging ideologies throughout LAC, this measure does not capture the

whole picture of intricate geopolitical relationships between exporters and importers. Second, the results should be interpreted as conditional correlations rather than definitive causal effects, since other time pair varying shocks may bias the results. Finally, the use of regional averages introduces a blurred description of the region since I am not able to distinguish between each country individually. However, despite these limitations, the robustness tests and the narrative help to bring balance and acknowledge the highly heterogenous effects.

4 Bayesian estimation

The Bayesian dynamic gravity model extends the static PPML framework by allowing bilateral trade flows to adjust gradually over time. The dependent variable is defined as $\ln(1 + \text{tradeflow}_{ijt})$, where i denotes the exporting country, j denotes the importing country, and t denotes year. The decision to include the logarithm is to alleviate the problem that zeroes present that PPML originally solved. The inclusion of the lagged dependent variable captures persistence in bilateral trade relationships, while country-pair fixed effects control for time-invariant bilateral characteristics such as distance, common language, historical ties, and other structural determinants of trade. Year fixed effects absorb common global shocks affecting all country pairs in a given year. The estimated model is a replica of the model presented in table 1 column (5) including the whole sample:

$$\ln(1 + \text{tradeflow}_{ijt}) = \rho \ln(1 + \text{tradeflow}_{ij,t-1}) + X_{ijt}\beta + \alpha_{ij} + \gamma_t + \varepsilon_{ijt}, \tag{3}$$

where ρ is the dynamic persistence parameter, X_{ijt} contains the geopolitical fragmentation interaction terms, α_{ij} represents country-pair fixed effects, γ_t represents year fixed effects, and ε_{ijt} is the error term. The parameter ρ measures the extent to which past bilateral trade predicts current bilateral trade after accounting for fixed effects and covariates. A positive

and statistically meaningful value of ρ implies that trade adjustments are gradual rather than instantaneous.

Prior to Bayesian estimation, the dependent variable, lagged dependent variable, and all covariates are residualized with respect to country-pair and year fixed effects. This applies the Frisch-Waugh-Lovell logic to remove the fixed effects from the estimating equation while preserving the identifying variation in the covariates of interest. The posterior estimation is then performed on the residualized outcome and residualized regressors.

The Bayesian model assumes a Gaussian likelihood for the residualized log trade equation. The coefficient vector, including the persistence parameter ρ and the fragmentation interaction coefficients, is assigned a weakly informative normal prior:

$$\theta \mid \sigma^2 \sim N(0, 100\sigma^2 I),$$

where $\theta = (\rho, \beta')'$ and I is an identity matrix. The error variance follows an inverse-gamma prior:

$$\sigma^2 \sim IG(2, 1).$$

These priors are intentionally diffuse and allow the posterior estimates to be driven primarily by the data rather than by strong prior assumptions since trade data is very complete.

Posterior simulation is carried out using a Gibbs sampler. Conditional on σ^2, the coefficient vector is drawn from its multivariate normal posterior distribution. Conditional on the coefficient vector, σ^2 is drawn from its inverse-gamma posterior distribution.

$$\theta \mid \sigma^2, y, X$$

and

$$\sigma^2 \mid \theta, y, X$$

This is a Gibbs sampler rather than a Metropolis-Hastings or Hamiltonian Monte Carlo algorithm, since each conditional posterior distribution has a known closed form distribution.

The model is estimated using four Markov chains. For each chain, 4,000 posterior iterations are generated after an initial burn-in period of 1,000 iterations. The chains are thinned by retaining every second draw, resulting in 2,000 retained posterior draws per chain and 8,000 retained posterior draws in total. Posterior inference is based on the retained draws. For each parameter, the reported posterior mean summarizes the central tendency of the posterior distribution, the posterior standard deviation measures posterior uncertainty, and the 2.5 and 97.5 percent quantiles form the 95 percent credible interval. The posterior probability $\Pr(\theta > 0)$ reports the share of retained posterior draws in which the parameter is positive.

The dynamic structure of the model implies that the estimated coefficients on the fragmentation interaction terms represent short-run effects. Because trade adjusts gradually over time, the long-run effect of each covariate is calculated as:

$$\frac{\beta}{1-\rho}$$

This transformation is computed draw-by-draw from the posterior distribution, rather than only from posterior means. Therefore, the reported long-run credible intervals incorporate uncertainty from both the short-run coefficient β and the persistence parameter ρ. This is important because the long-run effect depends directly on the degree of trade persistence estimated by the model.

Table 4: Bayesian Dynamic Gravity Model Estimates

Parameter	**Mean**	**SD**	**2.5%**	**97.5%**	**Pr($\theta > 0$)**
Panel A: Short-run posterior effects					
ρ	0.4291	0.0012	0.4268	0.4313	1.000
GFI × LAC intraregional	-0.1249	0.0465	-0.2165	-0.0299	0.003
GFI × LAC exporter	0.0246	0.0197	-0.0140	0.0656	0.903
GFI × LAC importer	0.0255	0.0204	-0.0140	0.0660	0.893
GFI × LAC intraregional × FTA	-0.0281	0.0852	-0.1942	0.1364	0.378
GFI × LAC exporter × FTA	0.1260	0.0515	0.0244	0.2253	0.993
GFI × LAC importer × FTA	-0.0344	0.0556	-0.1426	0.0768	0.273
GFI × FTA	-0.0151	0.0147	-0.0435	0.0145	0.151
Panel B: Long-run posterior effects, $\beta/(1-\rho)$					
LR: GFI × LAC intraregional	-0.2187	0.0814	-0.3796	-0.0522	0.003
LR: GFI × LAC exporter	0.0432	0.0346	-0.0245	0.1150	0.903
LR: GFI × LAC importer	0.0446	0.0357	-0.0245	0.1156	0.893
LR: GFI × LAC intraregional × FTA	-0.0492	0.1492	-0.3403	0.2393	0.378
LR: GFI × LAC exporter × FTA	0.2206	0.0902	0.0427	0.3948	0.993
LR: GFI × LAC importer × FTA	-0.0603	0.0974	-0.2496	0.1347	0.273
LR: GFI × FTA	-0.0265	0.0258	-0.0761	0.0253	0.151

Notes: The table reports posterior means, posterior standard deviations, 95 percent credible intervals, and posterior probabilities that the parameter is positive. The dependent variable is $\ln(1+\text{tradeflow}_{ijt})$. The model includes country-pair fixed effects and year fixed effects. ρ denotes the coefficient on the lagged dependent variable. Long-run effects are computed as $\beta/(1-\rho)$. GFI refers to the geopolitical fragmentation index from Fernández-Villaverde et al. (2024). LAC denotes Latin America and the Caribbean. FTA denotes the presence of a free trade agreement.

The posterior mean of ρ is 0.4291, with a 95 percent credible interval from 0.4268 to 0.4313 and a posterior probability of being positive equal to 1.000. This indicates a high degree of persistence in bilateral trade flows. Economically, approximately 43 percent of the previous period's trade level carries over into the current period, conditional on the fixed effects and the covariates included in the model. Since the model is dynamic, short-run coefficients do not fully capture the total adjustment effect. Long-run effects are therefore computed as $\beta/(1-\rho)$, which scales each short-run coefficient by the implied persistence of trade flows.

The coefficient on GFI × LAC intraregional trade is negative and credibly different from zero. The short-run posterior mean is -0.1249, with a 95 percent credible interval from -0.2165 to -0.0299 and a posterior probability of being positive equal to 0.003. This suggests that higher geopolitical fragmentation is associated with lower trade among LAC countries, relative to the omitted comparison group (Non LAC to non LAC partners). The corresponding long-run effect is -0.2187, with a credible interval from -0.3796 to -0.0522. This indicates that the negative impact of geopolitical fragmentation on intraregional LAC trade becomes stronger once dynamic adjustment is taken into account.

The coefficients on GFI × LAC exporter and GFI × LAC importer are positive in posterior mean, but their credible intervals include zero. The short-run posterior means are 0.0246 and 0.0255, respectively, with posterior probabilities of being positive equal to 0.903 and 0.893. These results provide suggestive, but not conclusive, evidence that trade between LAC and non-LAC partners may increase during periods of geopolitical fragmentation. However, because the credible intervals overlap zero, these effects should be interpreted cautiously.

The interaction between geopolitical fragmentation, LAC exporter status, and free trade agreements is positive and credibly different from zero. The short-run posterior mean of GFI × LAC exporter × FTA is 0.1260, with a 95 percent credible interval from 0.0244 to 0.2253 and a posterior probability of being positive equal to 0.993. The long-run effect

is 0.2206, with a credible interval from 0.0427 to 0.3948. This result suggests that LAC exporters trading with non-LAC partners under free trade agreements may benefit from geopolitical fragmentation, potentially through trade diversion, preferential market access, or the reallocation of supply chains toward countries perceived as alternative suppliers.

By contrast, the FTA interaction for intraregional LAC trade and the FTA interaction for non-LAC exporters selling to LAC importers are not credibly different from zero. Their credible intervals include both negative and positive values, indicating substantial posterior uncertainty. Similarly, the general GFI × FTA coefficient is slightly negative in posterior mean but not credibly different from zero. These findings suggest that the positive role of FTAs is not uniform across all trade relationships. Rather, the clearest evidence appears specifically for LAC exporters trading with non-LAC partners under free trade agreements. This is a key contrast to the previous models under standard frequentist theory. Policy recommendations no longer indicate a clear cut regional free trade agreement, there is more nuance. Free trade agreements may behave differently according to each trade relationship. This still should be interpreted carefully since there is not enough evidence to suggest drifting away from intraregional trade towards other agreements.

Table 5: Bayesian Dynamic Gravity Model Diagnostics

Parameter	**Mean**	**SD**	**2.5%**	**97.5%**	**Pr($\theta > 0$)**	$\hat{R}$	**ESS**	**MCSE**
ρ	0.429022	0.001166	0.426773	0.431355	1.00000	1.000205	3401.32	0.000020
GFI × LAC intraregional	-0.125257	0.048684	-0.220594	-0.028800	0.00375	1.000717	3976.08	0.000772
GFI × LAC exporter	0.024512	0.019466	-0.013798	0.062881	0.90025	0.999757	4000.00	0.000308
GFI × LAC importer	0.025450	0.019499	-0.012036	0.063910	0.89850	0.999928	3630.40	0.000324
GFI × LAC intraregional × FTA	-0.029041	0.086368	-0.196574	0.138434	0.37700	1.000043	3763.94	0.001408
GFI × LAC exporter × FTA	0.126121	0.052053	0.023710	0.228649	0.99425	0.999828	3785.52	0.000846
GFI × LAC importer × FTA	-0.035291	0.052955	-0.138706	0.068535	0.25200	0.999791	3866.02	0.000852
GFI × FTA	-0.015195	0.014660	-0.043558	0.014301	0.14650	1.000068	3870.66	0.000236
σ^2	1.398998	0.002567	1.394013	1.404154	1.00000	0.999760	3977.20	0.000041

Notes: The table reports posterior means, posterior standard deviations, 95 percent credible intervals, posterior probabilities that the parameter is positive, Gelman-Rubin $\hat{R}$ statistics, approximate effective sample sizes, and Monte Carlo standard errors. The model is estimated using a Gibbs sampler. GFI denotes the geopolitical fragmentation index. LAC denotes Latin America and the Caribbean. FTA denotes a free trade agreement.

Posterior convergence and simulation reliability are evaluated using trace plots, Gelman-Rubin $\hat{R}$ statistics, approximate effective sample sizes, and Monte Carlo standard errors. The diagnostic results indicate satisfactory convergence. The $\hat{R}$ values shown in table 5 are extremely close to one for all parameters, ranging approximately from 0.9998 to 1.0007. This suggests that the two chains converged to the same posterior region. The approximate effective sample sizes are also high, ranging from approximately 3,401 to 4,000 across the main coefficients. The Monte Carlo standard errors are very small relative to the posterior standard deviations, indicating that simulation error is negligible compared with posterior uncertainty.

The trace plots provide visual evidence of convergence for the most important parameters. Figure 5 shows the trace plot for the coefficient on the intraregional trade interaction. The two chains overlap closely and fluctuate around a stable negative posterior region, with no visible trend or systematic drift. This supports the reliability of the negative posterior estimate for intraregional LAC trade.

Figure 5

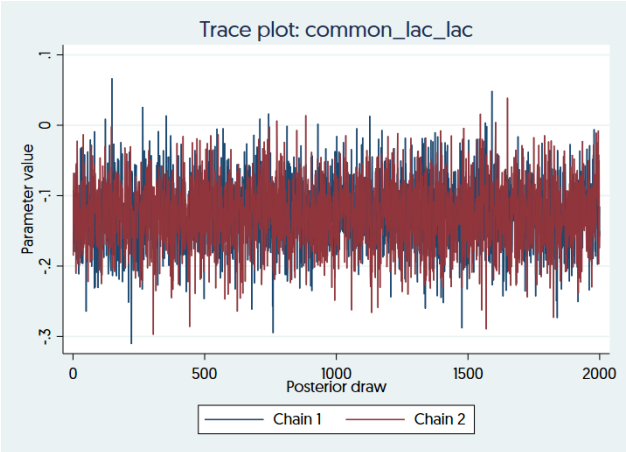


Figure 6 shows the trace plot for common_lac_nonlac_fta. The chains again overlap substantially and remain centered around a positive posterior region. This supports the interpretation that the positive effect for LAC exporters trading with non-LAC partners under free

trade agreements is not driven by poor chain mixing or lack of convergence.

Figure 6

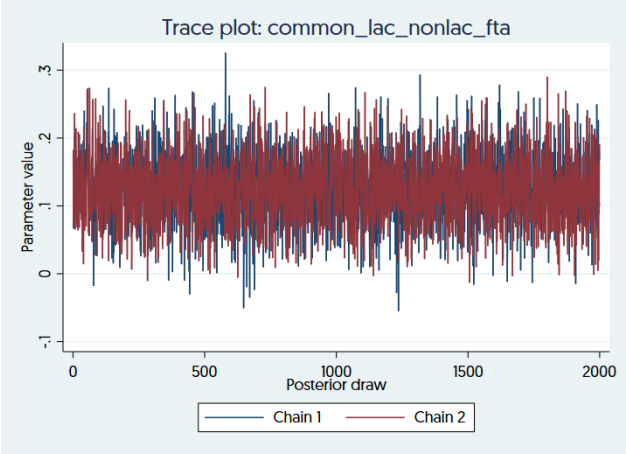
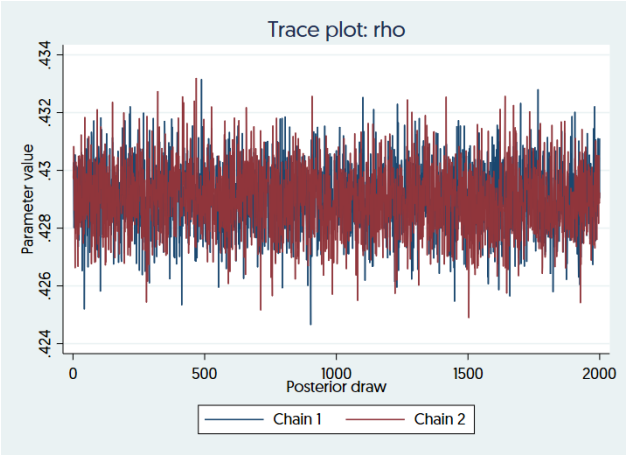


Figure 7 shows the trace plot for ρ, the dynamic persistence parameter. The chains are tightly concentrated around 0.429 and display stable mixing across the posterior draws. Since ρ determines the long-run multiplier, convergence of this parameter is particularly important. The stable trace plot and the $\hat{R}$ value of 1.0002 indicate that the estimated persistence parameter is reliable.

Figure 7



Overall, the Bayesian dynamic estimates suggest that geopolitical fragmentation has asymmetric effects across trade relationships. The strongest negative effect is observed for intrare-

gional LAC trade, while the strongest positive effect appears for LAC exports to non-LAC partners when an FTA is present. These findings indicate that geopolitical fragmentation may weaken regional trade integration within LAC while simultaneously creating external market opportunities for LAC exporters connected to non-LAC economies through formal trade agreements.

5 Conclusions

This essay analyses the effect of geoeconomic fragmentation on LAC's flows of trade. Using a gravity framework with country-pair and year fixed effects, the results show a consistent negative association between fragmentation measured by the index proposed by Fernández-Villaverde et al., 2024 and intraregional LAC trade. A 0.1 unit increase in the Geopolitical Fragmentation Index is associated with a decline of 2.9% in intra-LAC bilateral trade in the baseline specification, and this effect becomes stronger in the post-2008 period, reaching about 5.2%. These results are robust to other measures of index and trade flows. Moreover, they are a lower bound This suggests that *slowbalization* has not automatically strengthened regional trade linkages in LAC. Instead, fragmentation appears to weaken them.

The results also show that this effect is not uniform across countries or trade relationships. Free trade agreements partly mitigate the negative effect of fragmentation on intraregional trade, which suggests that institutionalized trade arrangements can reduce the vulnerability of regional flows to global geopolitical shocks. Using the SIE database I then extended the analysis to specifications which indicate that some LAC countries, especially those classified as more vulnerable through trade openness or terms-of-trade exposure, appear to redirect trade outside the region. The evidence on flows with China, the United States, and the European Union points toward trade diversion rather than a simple contraction of trade as

the baseline model suggested.

These findings challenge the idea LAC can only benefit from geopolitical realignment. While fragmentation may create opportunities for trade diversion, nearshoring, and stronger ties with major blocs, it also risks deepening LAC's dependence on external partners if intraregional trade weakens.

The main policy implications suggest regional integration must become the main target instead of trade diversion. This means reducing infrastructure gaps, harmonizing technical and regulatory barriers, strengthening trade facilitation, improving governance, and making free trade agreements more effective as suggested previously by Machado Parente (2024).

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